MINERVA INVESTMENT MANAGEMENT SOCIETY
Quantitative Research Division



Variance Risk Premia and Stock Return Predictability

Nonlinear Evidence from Polynomial and Spline Models

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Contents

1 Introduction **3**

2 Data **3**
2.1 Data Construction 3
2.2 Treatment of Missing Data 4
2.3 Summary Statistics 4
2.4 Volatility Regimes 5

3 Methodology **5**
3.1 Baseline Linear Specification 5
3.2 Polynomial Specification 6
3.3 Spline-Based Specification 6
3.4 Regime-Dependent Nonlinearities 6
3.5 Out-of-Sample Forecasting 6

4 Results **7**
4.1 Linear Benchmark 7
4.2 Polynomial Specification 7
4.3 Spline-Based Specification 7
4.4 Regime-Dependent Nonlinearities 8
4.5 Discussion 8

Abstract

This paper revisits the predictive power of the variance risk premium (VRP) for future stock returns, building on the seminal work of Bollerslev, Tauchen, and Zhou (2009). While prior studies primarily rely on linear predictive regressions, this paper investigates whether allowing for nonlinearities improves forecasting performance. Using U.S. equity data from 2008 to 2021, the analysis compares linear models with polynomial and spline-based specifications, as well as regime-dependent extensions based on volatility states.
The results show that linear models exhibit weak out-of-sample predictive power, while simple nonlinear transformations substantially improve forecasting performance. In particular, spline-based models deliver the strongest out-of-sample results, indicating that the relationship between the VRP and future returns is highly nonlinear. However, more complex regime-dependent models, despite improving in-sample fit, fail to outperform simpler specifications out of sample, suggesting overfitting in a limited sample environment.
Overall, the findings highlight the importance of nonlinear modeling for capturing the predictive content of the VRP, while emphasizing that parsimonious specifications provide the most robust and reliable forecasts in finite samples.

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1 Introduction

Predicting stock returns remains a central challenge in financial economics. Even though, since the beginning of arbitrage pricing theory, a vast number of predictors have been proposed, volatility-based measures have received particular attention due to their close connection to risk and investor expectations. Among these, the variance risk premium (VRP), which is defined as the difference between implied and realized volatility, has emerged as a key variable capturing compensation for bearing volatility risk. Prior research has shown that this indicator contains valuable information about future equity returns, making it a prominent predictor in both academic and applied settings.
A seminal contribution in this literature is provided by Bollerslev, Tauchen, and Zhou (2009) [1], who demonstrate that the spread between implied and realized volatility significantly predicts future stock returns. Their empirical analysis relies on linear predictive regressions, establishing a robust and economically meaningful relationship between the VRP and subsequent returns. However, the assumption of linearity may be restrictive. Financial markets are characterized by regime changes, asymmetric responses to shocks, and nonlinear risk-return trade-offs, suggesting that the predictive relationship between the VRP and returns may itself be nonlinear.
This paper revisits the predictive content of the variance risk premium by allowing for nonlinearities in a flexible but parsimonious way. Instead of relying on machine learning methods, the analysis employs polynomial specifications and spline-based regressions to capture potential nonlinear patterns in the data. Further variance-based regime assignments aim to enhance the models. These approaches retain interpretability while permitting richer functional forms than standard linear models. The main objective is to assess whether introducing nonlinearities improves the predictive performance of the VRP for future stock returns. The results indicate that accounting for nonlinear effects improves OOS forecasting and provides new insights into the nature of the risk-return relationship embedded in volatility markets.

2 Data

The empirical analysis combines data from multiple sources to construct the variance risk premium (VRP), excess stock returns, and a set of standard macro-financial control variables. The sample spans the period from 2008 to 2021 and is primarily driven by the availability of high-frequency S&P 500 data.

2.1 Data Construction

Excess stock returns are computed using S&P 500 index data in excess of the risk-free rate. The latter is proxied by Treasury bill yields obtained from the Federal Reserve Economic Data (FRED) database. Implied volatility is measured using the VIX index, also sourced from FRED, following the standard approach in the literature. [1]
Realized volatility is constructed from high-frequency intraday S&P 500 data obtained from Kaggle [1]. The original one-minute observations are aggregated to five-minute intervals to mitigate microstructure noise and to align with the methodology in Bollerslev et

[1] https://www.kaggle.com/datasets/gratefuldata/intraday-stock-data-1-min-sp-500-200821?

al. (2009) [1]. Realized variance is then computed as the sum of squared intraday returns over each period.

The variance risk premium is defined as:

$$VRP_t = IV_t - RV_t,$$

where IV_t denotes implied variance and RV_t realized variance.
In addition to the VRP, the analysis includes a set of standard predictive variables commonly used in the return predictability literature. Valuation ratios, including the log price-earnings ratio ($\log PE$) and log dividend-price ratio ($\log PD$), are obtained from Robert Shiller's dataset [3]. The consumption-wealth ratio (CAY) is taken from [2]. Further controls include the default spread (DFSP), the term spread (TMSP), and a relative risk-free rate measure (RREL).

2.2 Treatment of Missing Data

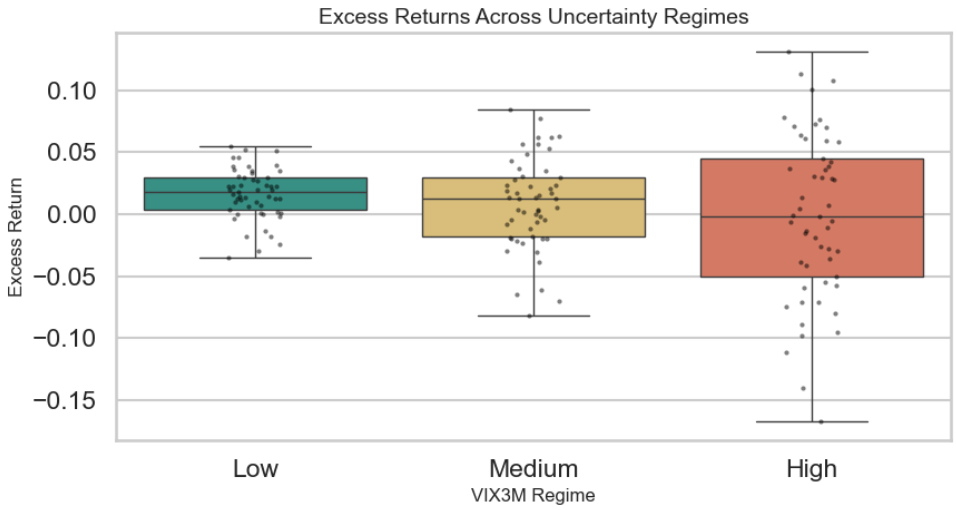
The dataset given its' sources was almost left without any missing. A practical challenge arises from the fact that the CAY series is only available until 2019, while the return and volatility data extend to 2021. To maintain a consistent sample, the missing observations for the final two years are imputed using a state-space Kalman model.
Specifically, CAY is modeled using an unobserved components model with a local level specification and macro-financial variables as exogenous predictors. The model is then estimated via maximum likelihood, and the fitted values are used to replace missing observations.
This approach allows for a smooth continuation of the CAY series while preserving its relationship with key economic variables.

2.3 Summary Statistics

Table 1: Summary Statistics

	Excess Ret	VRP	IV	RV	log PE	log PD	DFSP	TMSP	RREL	CAY
Mean	0.09	0.05	0.60	0.55	3.14	3.90	1.11	1.86	-0.13	-0.02
Std. dev.	0.16	0.12	0.21	0.12	0.43	0.14	0.50	0.97	0.50	0.02
Skewness	-0.66	3.80	3.70	3.56	2.26	-1.60	2.60	-0.17	-1.88	0.04
Kurtosis	1.61	18.88	15.89	17.29	5.72	5.13	7.79	-0.65	4.25	-0.56
AR(1)	0.08	0.76	0.76	0.59	0.98	0.95	0.95	0.98	0.96	0.95

Table 1 reports summary statistics for the main variables. Excess returns exhibit moderate volatility and slight negative skewness, consistent with standard findings. The VRP is highly persistent, with an AR(1) coefficient of 0.76, and displays pronounced skewness and excess kurtosis, reflecting the presence of extreme volatility episodes.
Both implied and realized volatility exhibit strong persistence and non-normality, consistent with volatility clustering in financial markets. Valuation ratios and macro variables also show high persistence, with AR(1) coefficients close to unity.



2.4 Volatility Regimes

To explore potential state dependence, the sample is partitioned into low, medium, and high volatility regimes based on the distribution of the VIX.
Figure 2.4 presents the distribution of excess returns across these regimes. The results reveal clear differences in return dynamics. Low-volatility periods are associated with relatively stable and positive returns. In contrast, high-volatility regimes exhibit substantially greater dispersion, more pronounced negative outcomes, and overall higher uncertainty. Medium regimes display intermediate characteristics but still show elevated variability compared to tranquil periods.

3 Methodology

This section describes the empirical framework used to evaluate the predictive power of the variance risk premium (VRP) for future stock returns. The analysis proceeds from a linear benchmark model similar to [1] to increasingly flexible nonlinear specifications, including polynomial regressions, spline-based models, and regime-dependent extensions. All models are estimated using heteroskedasticity and autocorrelation consistent (HAC) standard errors.

3.1 Baseline Linear Specification

As a benchmark, the predictive relationship between the VRP and future excess returns is estimated using the standard linear regression:

$$R_{t+1} = \alpha + \beta \cdot VRP_t + \varepsilon_{t+1},$$

where R_{t+1} denotes excess stock returns at time $t+1$, and VRP_t is the variance risk premium observed at time t. The model is estimated using ordinary least squares (OLS) with Newey-West (HAC) standard errors to account for potential serial correlation and heteroskedasticity.
To control for additional sources of return predictability, the specification is extended by including standard macro-financial predictors:

$$R_{t+1} = \alpha + \beta \cdot VRP_t + \gamma' X_t + \varepsilon_{t+1},$$

where X_t includes valuation and macro variables such as price-earnings ratios, term spreads, default spreads, and the consumption-wealth ratio.

3.2 Polynomial Specification

To allow for simple nonlinearities, the linear model is augmented with a quadratic term in the VRP:

$$R_{t+1} = \alpha + \beta_1 \cdot VRP_t + \beta_2 \cdot VRP_t^2 + \varepsilon_{t+1}.$$

This specification captures potential curvature in the relationship between the VRP and expected returns, allowing the marginal effect of the VRP to vary with its level.

3.3 Spline-Based Specification

To flexibly model nonlinearities without imposing a global functional form, spline regressions are employed. Specifically, cubic B-splines are used to approximate the relationship between VRP and future returns:

$$R_{t+1} = \alpha + \sum_{j=1}^{J} \beta_j B_j(VRP_t) + \varepsilon_{t+1},$$

where $B_j(\cdot)$ denotes spline basis functions. The spline specification uses a fixed number of degrees of freedom, allowing the relationship to vary smoothly across different regions of the VRP distribution. This approach captures local nonlinearities and potential threshold effects in a flexible yet parsimonious manner.

3.4 Regime-Dependent Nonlinearities

To account for potential state dependence, the spline specification is extended by interacting the spline basis functions with volatility regimes. Regimes are defined based on levels of three-month implied volatility (VIX3M), distinguishing between low, medium, and high volatility environments. The model takes the form:

$$R_{t+1} = \alpha + \sum_{j=1}^{J} \beta_j B_j(VRP_t) + \sum_{j=1}^{J} \beta_j^{(m)} B_j(VRP_t) \cdot D_t^{(m)} + \varepsilon_{t+1},$$

where $D_t^{(m)}$ are regime indicator variables. This specification allows the relationship between the VRP and expected returns to differ across volatility regimes, capturing both nonlinearities and state-dependent effects.

3.5 Out-of-Sample Forecasting

Finally, in order to evaluate predictive performance, all models are assessed using an out-of-sample (OOS) forecasting framework. Forecasts are generated using an expanding window approach, beginning with an initial estimation sample and recursively updating model parameters as new observations become available.

At each step, the model is re-estimated and used to produce one-step-ahead forecasts of excess returns. Predictive accuracy is evaluated using the root mean squared error (RMSE) and the out-of-sample R^2, defined relative to a historical mean benchmark.
This framework allows for a direct comparison of linear and nonlinear specifications in terms of their ability to improve return predictability.

4 Results

This section presents the empirical results, focusing on both in-sample fit and, more importantly, out-of-sample (OOS) predictive performance.

4.1 Linear Benchmark

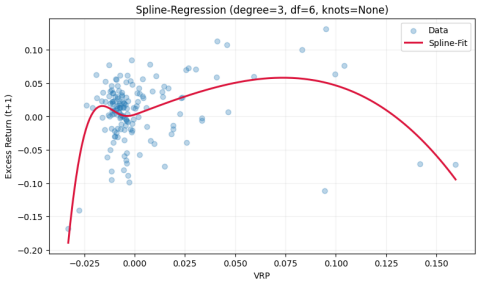
For the base linear model the coefficient on the VRP is positive but statistically insignificant, and the explanatory power is very low, with an R^2 of 0.015.
Out-of-sample performance is similarly weak. The linear model achieves an OOS R^2 of only 0.008, indicating that the VRP provides very limited predictive power in a purely linear specification in the examined time period. This result contrasts with earlier findings in the literature and suggests that the predictive relationship may not be well captured by a linear model in this sample.
Including macro-financial control variables improves the in-sample fit and renders the VRP coefficient statistically significant. However, as the focus of this paper is on parsimonious predictive structures, the subsequent analysis concentrates on nonlinear transformations of the VRP itself.

4.2 Polynomial Specification

Introducing a quadratic term in the VRP leads to a substantial improvement. Both the linear and squared terms are highly statistically significant, with opposite signs, indicating a concave relationship between the VRP and future returns.
Importantly, this nonlinear specification also improves out-of-sample performance. The polynomial model achieves an OOS R^2 of 0.086, representing a notable increase relative to the linear benchmark. This suggests that even simple nonlinear transformations are able to capture additional predictive information embedded in the VRP.

4.3 Spline-Based Specification

The spline-based model further enhances predictive performance by allowing for flexible, local nonlinearities. In-sample fit increases significantly, with an R^2 of 0.253, and most spline coefficients are highly statistically significant.
More importantly, the spline specification delivers the strongest out-of-sample performance among all models considered. The model achieves an OOS R^2 of 0.256, indicating a substantial improvement over both the linear and polynomial specifications. This result provides strong evidence that the relationship between the VRP and future returns is highly nonlinear and cannot be adequately captured by simple parametric forms.



4.4 Regime-Dependent Nonlinearities

Moreover, extending the spline model to allow for regime-dependent effects based on volatility states further increases the in-sample fit, with an R^2 of 0.332. This suggests considerable heterogeneity in the VRP-return relationship across different volatility environments.

However, this increased flexibility does not translate into superior out-of-sample performance. The regime-dependent spline model achieves an OOS R^2 of 0.176, which, while still positive and economically meaningful, is notably lower than that of the baseline spline model.

This divergence between in-sample and out-of-sample results suggests that the regime-dependent specification may suffer from overfitting. While volatility regimes clearly influence return dynamics 2.4 the additional model complexity appears to reduce forecasting robustness.

4.5 Discussion

Taken together, the results provide consistent evidence that nonlinearities play a crucial role in the predictive relationship between the variance risk premium and future stock returns. Moving from a linear to a nonlinear framework leads to substantial improvements in both in-sample fit and out-of-sample forecasting performance.

At the same time, the findings highlight an important trade-off between flexibility and robustness. While more complex models can better fit the data in-sample, simpler nonlinear specifications—such as spline regressions without regime interactions—deliver the strongest and most reliable out-of-sample performance.

A likely explanation for this pattern is the relatively limited sample size. The dataset comprises approximately 150 monthly observations, which restricts the ability to estimate highly flexible models without incurring substantial estimation error. In this setting, more complex specifications, such as regime-dependent splines, introduce a large number of parameters relative to the available data, increasing the risk of overfitting. As a result, these models may capture noise rather than stable underlying relationships, leading to weaker out-of-sample performance.

Overall, the results suggest that the predictive content of the VRP is inherently nonlinear, but that parsimonious nonlinear models provide the most effective and robust approach for capturing this relationship in finite samples.

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